Hyunwoo Shin

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Academic Appointment

MD Anderson Cancer Center — Houston, TX
Postdoctoral Fellow in Health Services Research — *Aug 2026 – Present*

- Conduct simulation-based analyses to evaluate and optimize lung cancer screening policies.
- Develop and validate lung cancer natural history microsimulation models to support screening policy analysis and research within the Cancer Intervention and Surveillance Modeling Network (CISNET).

Education

Virginia Polytechnic Institute and State University — Blacksburg, VA
Ph.D. in Industrial & Systems Engineering — *Aug 2021 – May 2026*

Korea Aerospace University — Goyang, South Korea
M.S. in Logistics, School of Air Transport, Transportation, and Logistics — *Mar 2018 – Feb 2020*

Korea Aerospace University — Goyang, South Korea
B.S. in Logistics, School of Air Transport, Transportation, and Logistics — *Mar 2012 – Feb 2018*

Publications

[1] **H. Shin**, S. Tunc, X. Chen, J. M. Reynolds. "A Comprehensive Simulation Framework for Evaluating U.S. Lung Allocation Policies." *Health Care Management Science*, 29(54), 2026. DOI: 10.1007/s10729-026-09797-z

[2] **H. Shin**, S. Tunc, X. Chen, J. M. Reynolds. "Forecasting Near-Term Changes in a Dynamic Allocation Score: Evidence from Lung Transplantation." *(under review at INFORMS Journal on Data Science)*

[3] S. Ishizawa, N. Collier, R. Nargund, M. Eghbalizarch, S. Mousavi, **H Shin**, E. Ostirn, S. Atkar-Khattra, E. Irajizad, J. Dennison, S. Hanash, M. Tammemagi, S. Lam, J. Ozik, R. Meza, I Toumazis. "Characterizing the Natural History of Lung Cancer by Sex, Histology, and Nodule attenuation: insights from the CHRONOS-Lung natural history model." *(under review at Chest)*

[4] **H Shin**, Y Ha "Multilevel Monte Carlo Trust-Region Method with Adaptive Sampling for Nested Simulation Optimization" *(Working paper)*

[5] **H. Shin**, S. Tunc. "Evidence-Based Policy Search under Simulation Uncertainty in Lung Allocation." *(Working paper)*

[6] Y. Ha, **H. Shin**, J. Mueller. "Scalable Traffic Signal Control for Urban Mobility: A Subspace-based Optimization Approach Using Deep Reinforcement Learning." *(Working paper)*

[7] **H. Shin**, and J. Chae. "A Performance Review of Collision-Free Path Planning Algorithms." *Electronics*, 9(2), 316, 2020. DOI: 10.3390/electronics9020316

[8] H.-Y. Lee, **H. Shin**, and J. Chae. "Path Planning for Mobile Agents Using a Genetic Algorithm with a Direction Guided Factor." *Electronics*, 7(10), 212, 2018. DOI: 10.3390/electronics7100212

[9] M. Kim, **H. Shin**, and J. Chae. "Merge Control using Reserve Ahead Point in Baggage Handling Systems." *Journal of the Society of Korea Industrial and Systems Engineering*, 40(2), 60–67, 2017.

Research Experience

National Renewable Energy Laboratory — Remote
Summer Intern (Mentor: Yunsoo Ha; Supervisor: Juliane Mueller) — *June 2025 – Aug 2025*

- Simulation optimization for traffic simulation.
- Finding the subspace for optimizing via the simulation model using a deep reinforcement learning model.

Virginia Tech Blacksburg, VA

Graduate Research Assistant (Advisor: Sait Tunc) *Spring 2022, Fall 2024, Fall 2025*

- Optimized U.S. lung allocation policy using high-performance computing and discrete-event simulation.
- Verified and validated a liver and kidney transplant simulation model via Bayesian optimization approaches and ASTRO-DF algorithm.
- Developed a simulation model for U.S. lung transplant allocation and processed large-scale clinical data.
- Investigated modeling techniques to estimate lung allocation scores under uncertainty.

Korea Aerospace University Goyang, South Korea

Researcher & Research Assistant (Advisor: Junjae Chae) *Mar 2016 – Jul 2021*

- Collaborated on a multi-shuttle control logic for an automated container yard (CyberLogitec & Korea Maritime Institute).
- Developed job assignment algorithms for a new business model (Nexen Tire & SL Solution).
- Analyzed automation logic for gantry cranes using simulation (Hanjin Logistics Institute).
- Evaluated methodologies for UAV path planning; conducted efficiency and performance analyses.
- Developed a vehicle routing problem (VRP) solution incorporating real map distances and operational constraints (SK Telecom & SL Solution).
- Built an unmanned logistics system with UAV pathfinding via Genetic Algorithms (Ministry of Land, Infrastructure and Transport).
- Analyzed a baggage handling system merging control logic using AutoMod.

Teaching Experience

Virginia Tech Blacksburg, VA

Instructor of Record, Department of Industrial & Systems Engineering *Spring 2026*

- **Undergraduate Course**: Logistics Engineering

Virginia Tech Blacksburg, VA

Teaching Assistant, Department of Industrial & Systems Engineering and Department of Statistics *2022 – 2025*

- **Undergraduate Courses (ISE)**: Deterministic Operations Research, Logistics Engineering, Data Management, Statistical Quality Control
- **Undergraduate Course (Statistics)**: Statistics for Engineers
- **Graduate Courses (ISE)**: Random Process, Facilities Planning

Korea Aerospace University Goyang, South Korea

Instructor of Record *Spring 2021*

- **Undergraduate Course**: Operations Research I

Conference Presentations

- **INFORMS Annual Meeting 2026**. **H. Shin**, X. Chen, and S. Tunc. "Evidence-Based Policy Search under Simulation Uncertainty in Lung Allocation."
- **INFORMS Annual Meeting 2025**. **H. Shin**, X. Chen, and S. Tunc. "Optimizing U.S. Lung Allocation Policy via a Multi-Objective Trust-Region-based Simulation Optimization Algorithm."
- **INFORMS Annual Meeting 2024**. **H. Shin**, X. Chen, and S. Tunc. "Calibration of Simulation Models for Organ Allocation Using Conformal Prediction Concepts."
- **INFORMS Annual Meeting 2023**. **H. Shin**, X. Chen, and S. Tunc. "Forecasting Organ Transplant Allocation Scores Using Machine Learning Models."

Honors & Awards

- **Paul E. Torgersen Graduate Student Research Excellence Award, Top Ten Finalist, Virginia Tech** 2026
- **ISE Graduate Student Travel Awards, Virginia Tech** 2023–2025
- **GPSS Travel Fund Program, Virginia Tech** 2025

Technical Skills

Languages: Python, C++, Julia, R, Java

Software Tools: AutoMod, Arena, ExtendSim, CPLEX, Gurobi

Methodologies: Simulation Optimization, Bayesian Optimization, Surrogate Modeling, Machine Learning, Deep Reinforcement Learning, Bayesian Inference

Operating Systems: Linux (Ubuntu/HPC), Windows (WSL)

Selected Coursework

- **Probability & Statistics:** ISE 5034 (*Math Probability & Statistics for ISEs*), ISE 5984 (*Stat Learning and Data Sci*), CS 5525 (*Data Analytics*), STAT 5114 (*Statistical Inference*), STAT 5444 (*Bayesian Statistics*), STAT 6105 (*Measure & Probability*), STAT 6474 (*Advanced Topics in Bayesian Statistics*), STAT 6544 (*Surrogate Modeling*)
- **Simulation & Stochastic Modeling:** ISE 5424 (*Simulation I*), ISE 6494 (*Advanced Simulation*), ISE 5414 (*Random Process*), ISE 6464 (*Queueing Networks*)
- **Optimization:** ISE 5405, 5406 (*Optimization I & II*)

Academic Service

- **Session Chair**: 2024, 2025, 2026 INFORMS Annual Meeting
- **VT INFORMS Student Chapter**: VP Event (2025–2026)